

Metrics for Evaluating Performance

1. Error (Basic)

- **Error** = Forecasted Value – Actual Value
 - Problem: Positive & negative errors can **cancel each other out**, giving a false sense of accuracy.
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2. Mean Squared Error (MSE)

- **Formula:** Mean of squared errors.
 - **Why square?** Removes negatives & penalizes large errors more.
 - **Use case:** When large mistakes are costly/dangerous.
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3. Root Mean Squared Error (RMSE)

- **Formula:** $\sqrt{\text{MSE}}$.
 - Same scale as original errors, still penalizes large errors more.
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4. Mean Absolute Error (MAE) (a.k.a. Mean Absolute Deviation – MAD)

- **Formula:** Mean of absolute errors.
 - Less sensitive to large errors than MSE.
 - **Use case:** When cost is proportional to error size.
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5. Mean Absolute Percentage Error (MAPE)

- **Formula:** $\text{Mean}(|\text{Error}| / |\text{Actual Value}|)$.
 - Measures error size **relative** to actual values (percentage).
 - Useful for comparing across different scales.
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➤ Key Insight:

- **Choose MSE/RMSE** → if big errors matter more.
 - **Choose MAE/MAD** → if all errors are equally costly.
 - **Choose MAPE** → if you want percentage-based comparison.
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